**MAT/MATH 215, Vize/Midterm, 19.11.2022, 09:30 - :00**

Name and Surname (İsim ve Soyisim) : _____

ID (Öğrenci No): : _____

Bölüm (Department) : _____

* This is a closed book and closed notes exam (Bu sınavda kitap ve not kullanılamaz).
* No calculators, no talking and no questions (Hesap makinası, konuşmak ve soru sormak yasaktır).

| 1 | 2 | 3 | 4 | 5 | Toplam |
|---|---|---|---|---|---|
| | | | | | |
| 0 | 0 | 0 | 0 | 0 | 100 |

**1.** Given that $y_1(t) = 1$ and $y_2(t) = \dfrac{1}{t}$ are solutions of the equation $t^2y'' + 2ty' = 0,\ t > 0$, find **a particular solution** of the equation

$$t^2y'' + 2ty' = \frac{1}{t},\ t > 0.$$

( $t^2y'' + 2ty' = 0,\ t > 0$, denkleminin çözümleri $y_1(t) = 1$ ve $y_2(t) = \dfrac{1}{t}$, olduğuna göre,

$$t^2y'' + 2ty' = \frac{1}{t},\ t > 0.$$

denkleminin **özel bir çözümünü** bulunuz.)

Solution: We will use the method of variation of parameters. First, we'll write the differential equation in standard form by dividing both sides of the equation by $t^2$:

$$y'' + \frac{2}{t}y' = \frac{1}{t^3},\ t > 0.$$

Then, a particular solution is $y_p = c_1(t) + c_2(t)\dfrac{1}{t}$ and $W[y_1, y_2] = A = det\begin{bmatrix} 1 & \frac{1}{t} \\ 0 & -\frac{1}{t^2} \end{bmatrix} = \dfrac{-1}{t^2}$

$$\begin{bmatrix} 1 & \frac{1}{t} \\ 0 & -\frac{1}{t^2} \end{bmatrix} \begin{bmatrix} c_1' \\ c_2' \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{1}{t^3} \end{bmatrix}$$

$$c_1' = \frac{\begin{vmatrix} 0 & \frac{1}{t} \\ \frac{1}{t^3} & -\frac{1}{t^2} \end{vmatrix}}{\begin{vmatrix} 1 & \frac{1}{t} \\ 0 & -\frac{1}{t^2} \end{vmatrix}} = \frac{-\frac{1}{t^4}}{\frac{-1}{t^2}} \Rightarrow c_1' = \frac{1}{t^2} \Rightarrow c_1 = -\frac{1}{t}$$

$$c_2' = \frac{\begin{vmatrix} 1 & 0 \\ 0 & \frac{1}{t^3} \end{vmatrix}}{\begin{vmatrix} 1 & \frac{1}{t} \\ 0 & -\frac{1}{t^2} \end{vmatrix}} = \frac{\frac{1}{t^3}}{\frac{-1}{t^2}} \Rightarrow c_2' = \frac{-1}{t} \Rightarrow c_2 = -\ln t.$$

Therefore,

$$y_p = c_1(t) + c_2(t)\frac{1}{t} = -\frac{1}{t} - \frac{1}{t}\ln t$$

is a particular solution.

**2.** The characteristic eqution associted to the given differential equation is $r^4 - 8r$, so the roots are $0, 2, -1 + \sqrt{3}i, -1 - \sqrt{3}i$. So $\{1, e^{2x}, e^{-x}\cos(\sqrt{3}x), e^{-x}\sin(\sqrt{3}x)\}$ is a fundamental set of solutions. Hence the general solution is as follows:

$$y(x) = C_1 + C_2e^{2x} + C_3e^{-x}\cos(\sqrt{3}x) + C_4e^{-x}\sin(\sqrt{3}x)$$

**3.** Find the general solution of following differential equation, given a particular solution $y_1(x) = \frac{1}{x}$

$$x^2y' + xy - x^2y^2 + 1 = 0$$

Özel çözümü $y_1(x) = \frac{1}{x}$ olarak verilen yukarıdaki diferansiyal denklemin genel çözümünü bulunuz.

**Solution:** Divide both side by $x^2$, then we have $y' + -\frac{1}{x^2} + z'y - y^2 + \frac{1}{x^2} = 0 \Rightarrow y' + \frac{1}{x}y - y^2 = -\frac{1}{x^2}$

Riccati D.E. with $a(x) = \frac{1}{x}, b(x) = -1$, and $c(x) = -\frac{1}{x^2}$

$y(x) = y_1(x) + z(x) = \frac{1}{x} + z(x) \Rightarrow y' = -\frac{1}{x^2} + z'$ plug this in above D.E

$$\begin{aligned} -\frac{1}{x^2} + z' + \frac{1}{x}\left(\frac{1}{x} + z\right) - \left(\frac{1}{x} + z\right)^2 &= -\frac{1}{x^2} \Rightarrow \\ z' - \frac{1}{x}z - z^2 &= 0 \text{ Bernoulli D. E.} \end{aligned}$$

$u = z^{-1} \Rightarrow u' = -\frac{1}{z^2}z'$ and multiply both side by $z^{-2}$ above Bernoulli D.E., then

$$u' + \frac{1}{x}u + 1 = 0 \text{ Linear D.E. with } a(x) = \frac{1}{x} \text{ and } b(x) = -1$$

Integrating multiple $e^{\int \frac{1}{x}dx} = e^{\ln|x|} = x$ Multiply linear D.E. with this intergrating multiple

$$\begin{aligned} xu' + u &= -x \\ (xu)' &= -x \end{aligned}$$

$$\begin{aligned} u &= -\frac{1}{2}x + \frac{C_1}{x} \Rightarrow z = \frac{2x}{C - x^2},\ C = 2C_1 \\ y(x) &= \frac{1}{x} + \frac{2x}{C - x^2} \end{aligned}$$

**4.**Find the solution of the given initial value problem.

$$\begin{cases} xy' + y(x^2 + \ln y) = 0 \\ y(1) = 1 \end{cases}$$

(Yukarıdaki verilen başlangıç değer probleminin çözümünü bulunuz.)

Solution: $xy' + y(x^2 + \ln y) = 0 \Rightarrow y(x^2 + \ln y)dx + xdy = 0.$

$P = y(x^2 + \ln y)$ and $Q = x \Rightarrow P_y = x^2 + \ln y + y\frac{1}{y} = x^2 + \ln y + 1$ , $Q_x = 1.$

$\frac{d\lambda}{\lambda} - \frac{Q_x - P_y}{P}dy = -\frac{1-x^2-\ln y-1}{y(x^2+\ln y)}dy = -\frac{1}{y}dy.$

$\lambda = \frac{1}{y} \Rightarrow \frac{1}{y}(y(x^2 + \ln y)dx + xdy) = 0$

$\Rightarrow (x^2 + \ln y)dx + \frac{x}{y}dy = 0.$ Note that $\frac{d}{dy}(x^2 + \ln y) = \frac{1}{y}$ and $\frac{d}{dx}(\frac{x}{y}) = \frac{1}{y}$.
So the new equation represents the exactly differential equation. let the solution of it be $\pi = C_1$. Then $\pi_x = x^2 + \ln y \Rightarrow \pi = \frac{x^3}{3} + x \ln y + f(y).$

$\pi_y = \frac{x}{y} + f'(y) = \frac{x}{y} \;\Rightarrow\; f'(y) = 0 \;\Rightarrow f(y) = C_0.$

$\frac{x^3}{3} + x \ln y = C$ where $C = C_1 - C_0$. Since $y(1) = 1$, then $\frac{1^3}{3} + 1 \ln 1 = C$ holds, that is, $C = \frac{1}{3}$, so the solution is $\frac{x^3}{3} + x \ln y = \frac{1}{3}$

**5.** Aşağıdaki başlangıç değer probleminin çözümüne 2. Picard yaklaşımını bulunuz.
*(Find the 2. Picard approxsimation to the solution of the following initial value problem)*:

$$\begin{cases} y' = \dfrac{3x^2+4x+2}{2(y-1)}, \\ \\ y(0) = 0. \end{cases}$$

Başlangıç yaklaşımı olarak $y_0(x) = 0$ alınız.
*(Take $y_0(x) = 0$ for intial approximation).*

**Solution:**

$$y' = \frac{3x^2+4x+2}{2(y-1)} = f(x,y)$$

$$y_0(x) = 0$$

$$y_n(x) = y_0(x) + \int_0^x f(s, y_{n-1}(s))\,ds \qquad (5)$$

$$y_0(x) = 0$$

**n = 1**

$$y_1(x) = \int_0^x \frac{3s^2+4s+2}{2(y_0(s)-1)}\,ds = \int_0^x \left(-\frac{3}{2}s^2 - 2s - 1\right)ds \qquad (3)$$

$$= \left(-\frac{s^3}{2} - s^2 - s\right)\Big|_0^x$$

$$\boxed{y_1(x) = -\frac{x^3}{2} - x^2 - x} \qquad (3)$$

**n = 2**

$$y_2(x) = \int_0^x \frac{3s^2+4s+2}{2\left(-\frac{s^3}{2} - s^2 - s - 1\right)}\,ds = \int_0^x \frac{3s^2+4s+2}{-s^3-2s^2-2s-2}\,ds$$

$y_1(s)$

$\{-s^3-2s^2-2s-2 = u,\ (-3s^2-4s-2)ds = du\}$

$$-\int \frac{du}{u} = -\ln|u| \qquad (3)$$

$$= -\ln|-s^3-2s^2-2s-2|\Big|_0^x \qquad (3)$$

$$\rightarrow y_2(x) = \ln 2 - \ln|-x^3-2x^2-2x-2| = \ln\left|\frac{2}{-x^3-2x^2-2x-2}\right| \qquad (3)$$

$$\ln\left|\frac{2}{x^3+2x^2+2x+2}\right|$$